Emil Kim

Busan, South Korea | emilkim01@pusan.ac.kr | +82 10-2892-7911
linkedin.com/in/emeelkaa | emeelkaa.github.io

Research Interests

Representation Learning • Multimodal Learning • Biomedical Signal Analysis

Education

Pusan National University — Busan, South Korea
M.Sc. in Artificial Intelligence, Expected Feb. 2027 — Mar 2025 – Present
GPA: 4.25/4.50

Pusan National University — Busan, South Korea
B.Sc. in Computer Science and Engineering — Sep 2020 – Feb 2025
GPA: 4.01/4.50 • Outstanding Graduate (Top 10 in Class)

Research Experience

Image Computing & Machine Learning Lab, Pusan National University — Busan, South Korea
Research Assistant — Sep 2024 – Present

- Developed a multimodal deep learning framework for EEG-to-fMRI synthesis, improving reconstruction quality while reducing model parameters by 3.2× and peak GPU memory usage by 2.3×.
- Designed and evaluated a hybrid Mamba–Transformer architecture for spatio-temporal EEG representation learning, improving balanced accuracy by 6.6 percentage points and reducing inference latency by 7.1% across three EEG datasets.
- Built a lightweight CNN–Transformer model for EEG seizure detection, achieving 99.55% accuracy on the CHB-MIT dataset with 238K parameters.

Ziyatron AI — Almaty, Kazakhstan
AI Researcher — Jan 2025 – Nov 2025

- Curated an EEG dataset from 92 pediatric patients in collaboration with the 2nd Pediatric Hospital and Kazakh Medical University in Almaty.
- Developed and integrated an EEG-based seizure detection model with GPT-4o for an AI-assisted clinical decision support platform.
- Presented the project at Digital Bridge 2025 in Astana, Kazakhstan.

Publications

Conference Proceedings

- E. Kim and J. K. Gahm, "MARBLE: Lightweight EEG-to-fMRI Translation via Mamba-Attention and ROI-Conditioned Decoding," MICCAI, Strasbourg, France, Sep 2026. **Paper** • **Code** • **Demo**
- E. Kim and J. K. Gahm, "CGM-EEG: Cross-Gated Mamba for Spatio-Temporal EEG Representation Learning," IEEE ISBI (Oral), London, UK, Apr 2026. **Paper** • **Code**

Journal Articles

- E. Kim and J. K. Gahm, "HSST-EEG: A Hybrid State-Space and Transformer Architecture for EEG Decoding," IEEE Access, vol. 14, pp. 79188–79196, May 2026. **Paper** • **Code**
- E. Kim, Y. Kenes, S. D. Tran, and J. K. Gahm, "Multi-Scale EEG Feature Extraction with Attention Mechanism for Seizure Detection," Journal of KIISE, vol. 53, no. 6, pp. 497–506, 2026. **Paper**

Teaching Experience

Teaching Assistant, Pusan National University — Busan, South Korea
Introduction to Unix Programming — Mar 2026 – Jun 2026

- Developed programming assignments, facilitated weekly lab sessions, and provided technical guidance to undergraduate students.

Awards & Scholarships

1. **1st Place, Three Minute Thesis (3MT)**, Pusan National University — Aug 2026
   - Won first place among 10 finalists from various departments, presenting EEG-to-fMRI research to a non-specialist audience.
2. **Best Paper Award**, Physical AI International Forum, Korean Artificial Intelligence Association — Nov 2025
   - Recognized for *HSST-EEG: A Hybrid State-Space and Transformer Architecture for EEG Decoding.*
3. **Graduate School Scholarship** (Full Tuition), Pusan National University — Mar 2025 – Present
4. **1st Place, PNU Ideathon**, Pusan National University — Dec 2024
   - Presented an EEG-based seizure detection model in a team competition among computer science students.
5. **Samsung Dream Scholarship**, Samsung Dream Scholarship Foundation — Sep 2023 – Dec 2024
   - Selected as one of two recipients per university for outstanding international students.
6. **Special Admission Scholarship** (Full Tuition), Pusan National University — Sep 2020 – Feb 2025

Poster Presentations

- E. Kim and J. K. Gahm, "Band-wise Variational EEG Representation Learning for Alzheimer's Disease Classification," Korean Artificial Intelligence Association Summer Conference (한국인공지능학회 하계학술대회), Gyeongju, South Korea, Aug 2026.
- E. Kim and J. K. Gahm, "HSST-EEG: A Hybrid State-Space and Transformer Architecture for EEG Decoding," Physical AI International Forum (한국인공지능학회 추계학술대회), Seoul, South Korea, Nov 2025.
- E. Kim, Y. Kenes, S. D. Tran, and J. K. Gahm, "Multi-Scale Attention Network for EEG-based Seizure Detection," Korea Computer Congress (한국컴퓨터종합학술대회), Jeju, South Korea, July 2025.

Skills

Programming Python, C, MATLAB, SQL
ML Frameworks PyTorch, Scikit-learn
Data Processing NumPy, Pandas, SciPy, MNE-Python
Tools & Platforms Git, Linux, Docker
Languages Russian (Native), English (IELTS 8.0), Korean (TOPIK 5)